Smart Inventory and Sales Analytics System

User Manual

Submitted by: Team Metaminds

Members:

Name	Class	Roll Number
Arpit Verma	CSE-B1	24293916006
Piyush Jayswal	CSE-B1	24293916017
Rahul Kumar	CSE-B1	24293916038
Dheeraj	CSE-B1	24293916043
Atharv Sinha	CSE-B1	24293916053

❖ Introduction:

The Smart Inventory and Sales Analytics System is a web-based platform designed to assist small businesses in managing inventory, billing, and sales analytics efficiently. The system reduces manual errors, improves operational speed, and provides actionable insights for better decision-making.

❖ System Requirements:

Hardware:
- Minimum 4GB RAM
- 500MB storage

Software:
- Python 3.10+
- MySQL Server
- Web Browser

❖ Installation and Setup:

1. Clone repository
   git clone https://github.com/Atharv2208Sinha/MetaMinds_SE_Project_CSEB1.git
   Setup environment
   python -m venv .venv
   source .venv/bin/activate
2. Install dependencies
   pip install -r requirement.txt
3. Configure database and run:
   python app.py

❖ Test Credentials:

Retail: Email – a@gmail.com; Password – As2208##
Pharmacist: Email – d@gmail.com; Password – As2208##

❖ Step-by-step Instructions:

1. Headers:

Fig – User not logged in

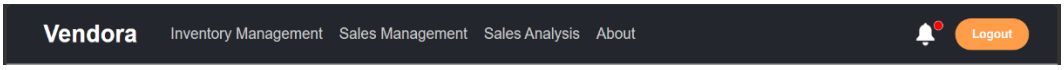


Fig – User logged in

Note:

1. Click on **'Vendora'** to go to **'Main'**.
2. Click on **Logout** button to log out and redirect to Home/About
3. Click on **Bell** icon to open **Notification** Modal. Bell icon will have red dot if there are any unread notifications.

2. Sign Up Page:

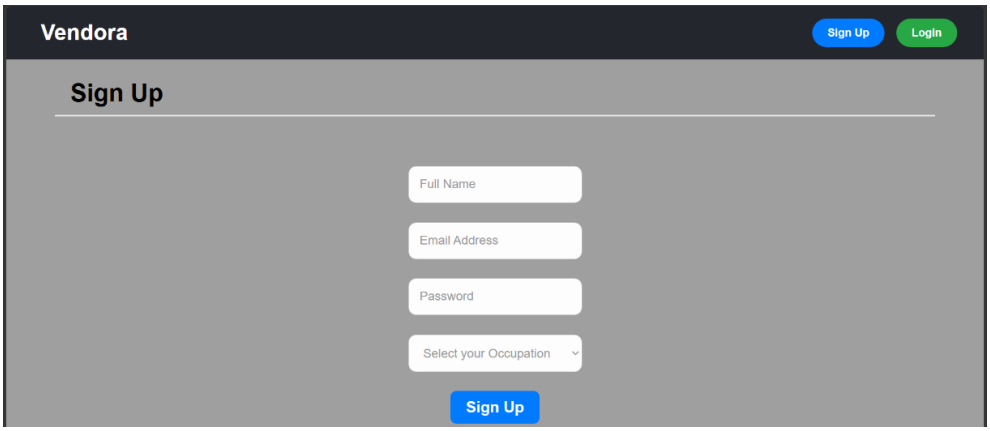


Fig – Sign Up Page

Purpose: To sign up new users.

Input Fields:

- **Alphanumeric Text –** Full Name; Email Address; Password
- **Selection Box –** Occupation

Instructions:

1. All fields are required. Fill name and email id.
2. Choose and fill password. Password must have 8+ characters, including uppercase, lowercase, number & special character.
3. Select occupation from options – Retail Shopkeeper; Pharmacist
4. Press **Sign Up** button. It will sign user up and redirect to Main.

3. Login Modal:

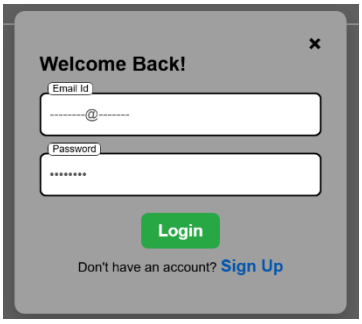


Fig – Login Modal

4. Main Page:

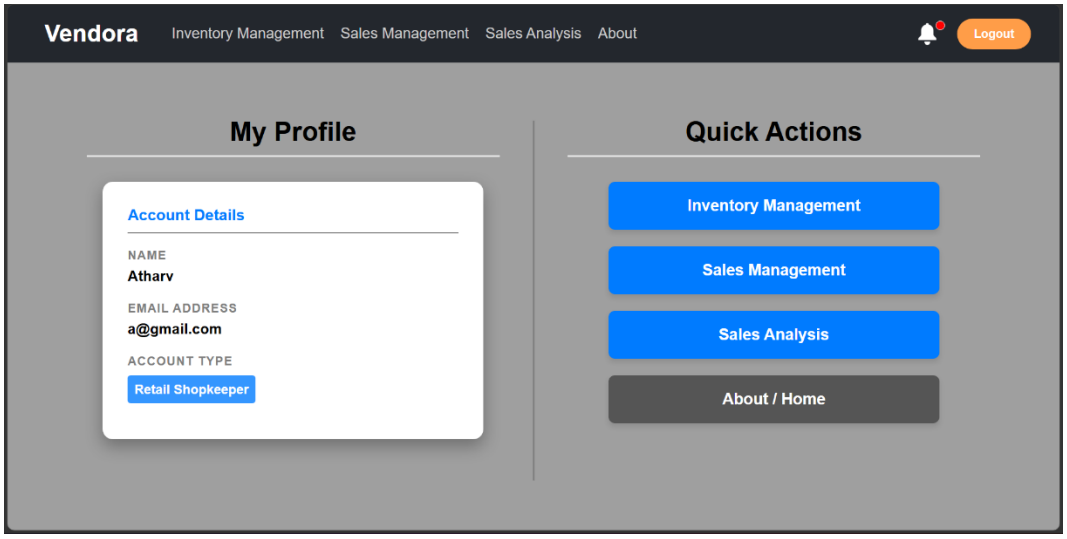


Fig – Main Page

Purpose: Profile/Navigation Page.

Input Fields: N/A

Instructions:

1. See account details from My Profile Section
2. Click on a button from Quick Actions Section to use that feature.

5. Notification Modal:

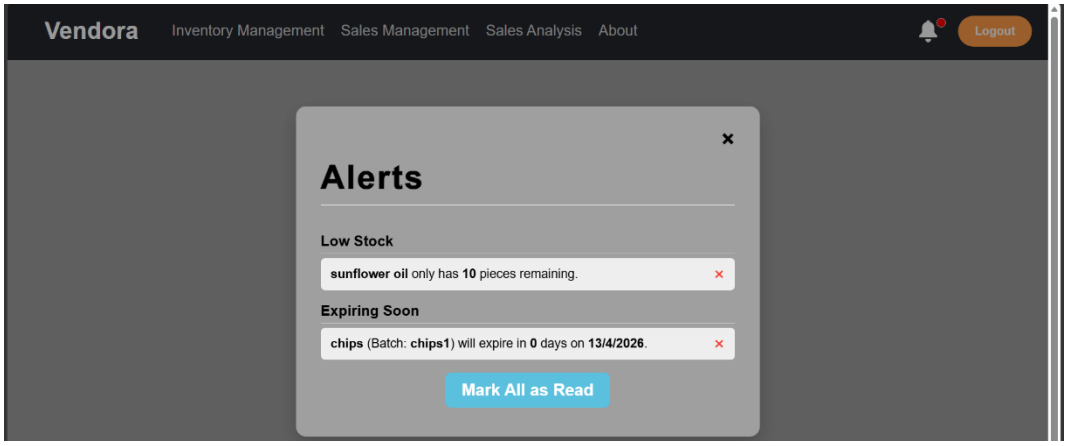


Fig – Notification Modal

Purpose: View Notifications.

Input Fields: N/A

Instructions:

1. Click on red cross on individual notifications to mark them as read.

2. Click on **Mark All as Read** button to mark all notifications as read.

Note: Daily notifications are sent for products that are running low or are about to expire. For stale products there is a gap of one month between consecutive notifications for that product.

6. Inventory Management Page:

➢ Feature: Add/Update Inventory

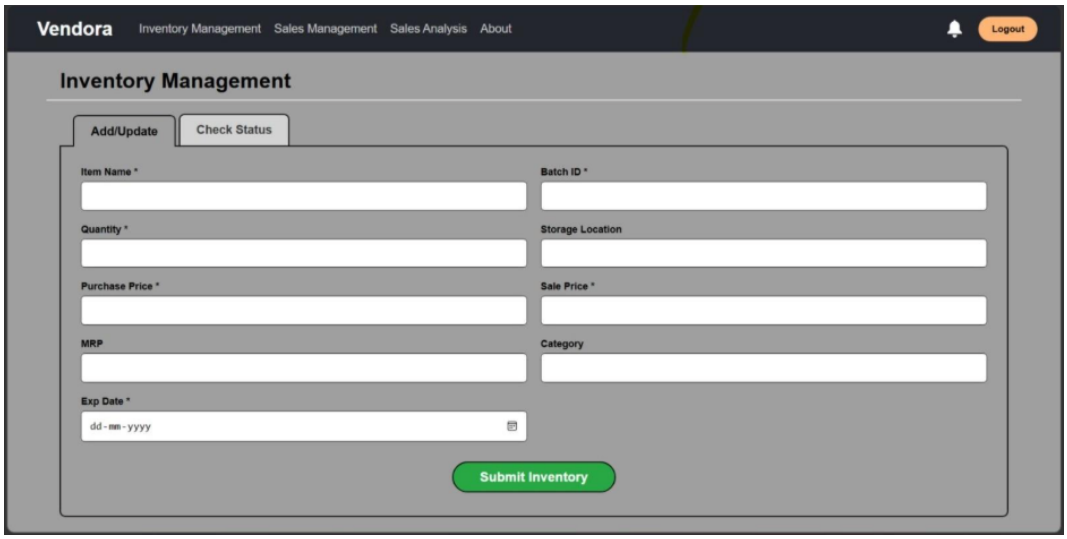


Fig: Add/Update Inventory Tab

Purpose: The **Add/Update** tab allows you to register new products or update existing stock levels in the Vendora database. Because Vendora serves a variety of small businesses, this page adapts based on your assigned user profile.

User Profiles & Interface Variations: Depending on your business type, you will see one of two interfaces:

1. **Standard Retail Vendor**

   Profile: Users managing general merchandise, electronics, or everyday retail items.

   Interface: You will see the standard inventory form focused on pricing, quantities, and basic categorization.

2. **Specialized / Pharmacy Vendor**

   Profile: Users managing medical supplies, supplements, or complex items where ingredient tracking is legally or operationally required.

   Interface: You will see the advanced inventory form. This includes an operation **Mode** dropdown at the top right and a dynamic **Composition** builder at the bottom of the form.

Instructions:

Part 1: Entering Basic Details (All Users)

- **Item Name:** The registered name of the product. Provides auto complete suggestion
- **Batch ID:** The unique identifier assigned to this specific stock batch. Displays all currently used batch ids beginning with the entered charcters as the user if filling it.
- **Quantity:** The current number of units available in this batch.
- **Purchase Price & Sale Price:** The cost to acquire the unit and the price it is sold to customers.
- **MRP:** The Maximum Retail Price of the item.
- **Exp Date:** The expiration date for this specific batch.
- **Purchase Date:** The date this batch was originally added to the system.
- **Category:** The product group the item belongs to (e.g., "instant food"). Provides autocomplete suggestion
- **Location:** The physical storage location of this batch (e.g., "A1").

Note: Fields with asterisk (*) are mandatory.

Part 2: Advanced Features (Pharmcists Only)

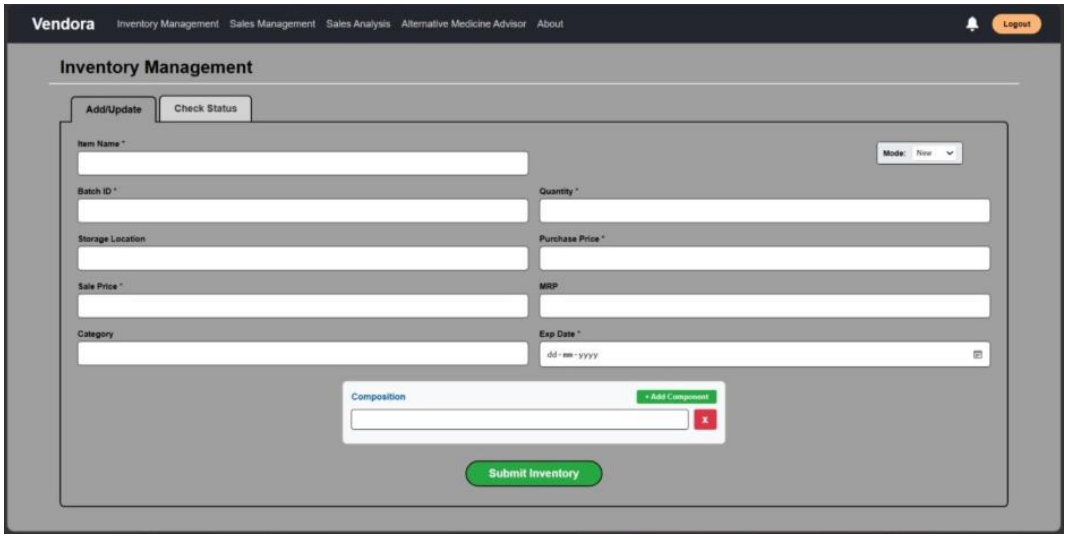


Fig – Inventory Management Page (Pharmacists)

If you are logged in as a Pharmacist, you must complete these additional steps:

1. **Selecting the Operation Mode:** Before typing, look at the **Mode** dropdown in the top right corner.
   i. Select **New** to register a completely new product batch into the database.
   ii. Select **Update** to modify the details of a batch that already exists in the system.
2. **Building the Product Composition:** Use the Composition section at the bottom to log specific active ingredients, chemicals, or materials.

i. **Add an Ingredient:** Type the name of the ingredient into the box. Choose from autocomplete suggestions.
ii. **Add Multiple Ingredients:** Click **+ Add Component** again for every additional ingredient you need to log.
iii. **Remove an Ingredient:** If you make a mistake, click the red **"X"** button next to the specific text box to delete that component.

Part 3: Submitting the Entry (All Users)

Once all applicable fields and components are filled out accurately, click the green **Submit Inventory** button at the bottom of the screen. The system will save your data and show a success message. You can verify the entry by navigating to the **Check Status** tab.

➢ Feature: Check Status

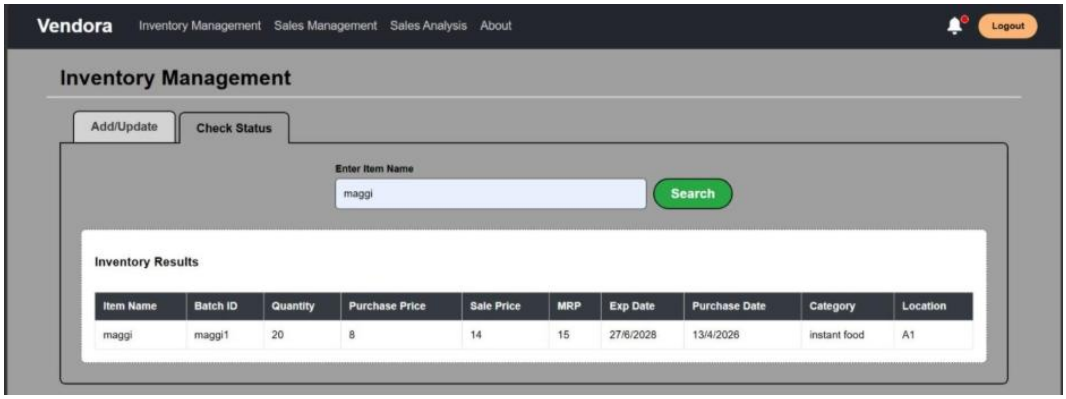


Fig: Check Status Tab

Purpose: The **Check Status** tab allows you to quickly search the database and view real-time details of items currently in your inventory. This feature is essential for verifying stock levels, checking expiration dates, and confirming that newly added items were successfully saved to the system.

Instructions:

1. **Search for an Item:** Locate the **Enter Item Name** search bar. Type the name of the product you want to find (for example, "maggi"). And
2. **Execute Search:** Click the green **Search** button. The system will query the database for all records matching that item name.
3. Once the search is complete, a table will appear displaying the fetched data. Each row represents a specific batch of the searched item.

Troubleshooting: If the table is empty after clicking Search, ensure the item name is spelled correctly or verify that the item has been added through the **Add/Update** tab first.

7. Sales Management Page

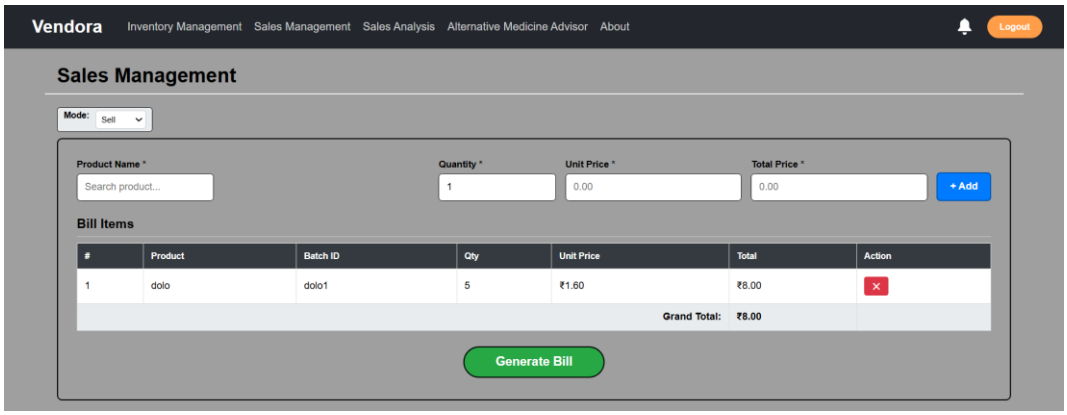


Fig – Sales Management Page

Purpose: To create and manage sales bills/invoices by adding products from inventory, specifying quantities, and generating printable invoices. Also handles removing Discarded products from inventory

Input Fields:

- **Mode –** Select if item is being sold or discarded
- **Product Name –** Search and select products from inventory
- **Quantity –** Numeric input for quantity to sell
- **Unit Price –** Price of single piece of items being sold
- **Total Price –** Total price of items being sold

Instructions:

1. **Select Mode:** Choose 'Sell' or 'Discard' from the selection box.
2. **Search and Select Product:** Enter Product name and select the batch from the autocomplete suggestions. On selecting its usual sales price, MRP and quantity will be displayed.
3. **Add Quantity**
4. **Set Price:** The input form will be automatically populated based on quantity and usual sales price(from the database). In case you want to give discount, edit the value with the final price you have in mind.
5. **Add item to Bill:** Click the '+ Add' button (blue button on the right). The product will be added to the Bill Items table below. The bill will auto calculate the price per piece.
6. **Review Bill Items:** All added items appear in the Bill Items table. The Grand Total is displayed at the bottom right, showing the sum of all line items.
7. **Remove item from Bill:** To remove an item, click the red '×' button in the Action column.
8. **Add More Products (Optional):** Repeat steps 2-5 to add multiple products to the same bill. You can add same products with different batch IDs to a single invoice.

9. **Generate Bill/Invoice:** Once all items are added, click the Generate Bill button. Upon generation a success message will appear: "Bill generated successfully!"
10. **View/Print Invoice:** After generating, a PDF invoice is automatically created and can be viewed/downloaded.

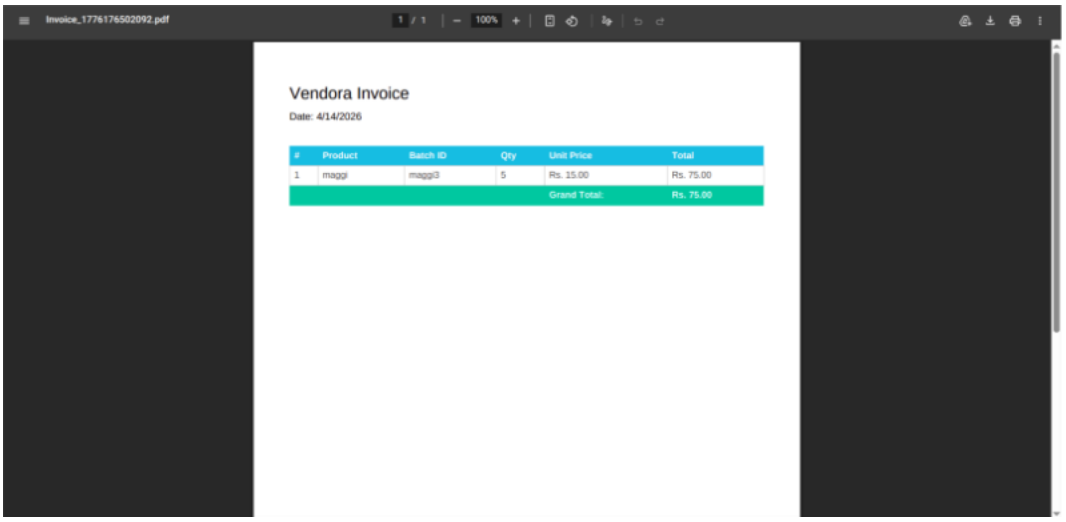


Fig – Generated Invoice

Note:

- **Stock Validation:** The system will not allow you to add quantities exceeding available stock for a specific batch.
- **Batch Selection:** Products are tracked by batch ID. If multiple batches exist for the same product, you must select the specific batch you want to sell from.
- **Price Modification:** While the price auto-populates from the product's sale price, you can manually adjust both unit price and total price if needed(e.g., for discounts/ price hikes). The price gets updated if either quantity, unit price or total price is modified.
- **Bill Clearing:** After successfully generating a bill, the Bill Items table is cleared and ready for the next transaction.
- **Invoice Numbering:** Each generated invoice receives a unique identifier (visible in the PDF filename, e.g., Invoice_1776176502092.pdf).
- **Inventory Update:** When a bill is successfully generated, the inventory stock is automatically reduced by the sold quantities.

8. Alternative Medicine Advisor Page

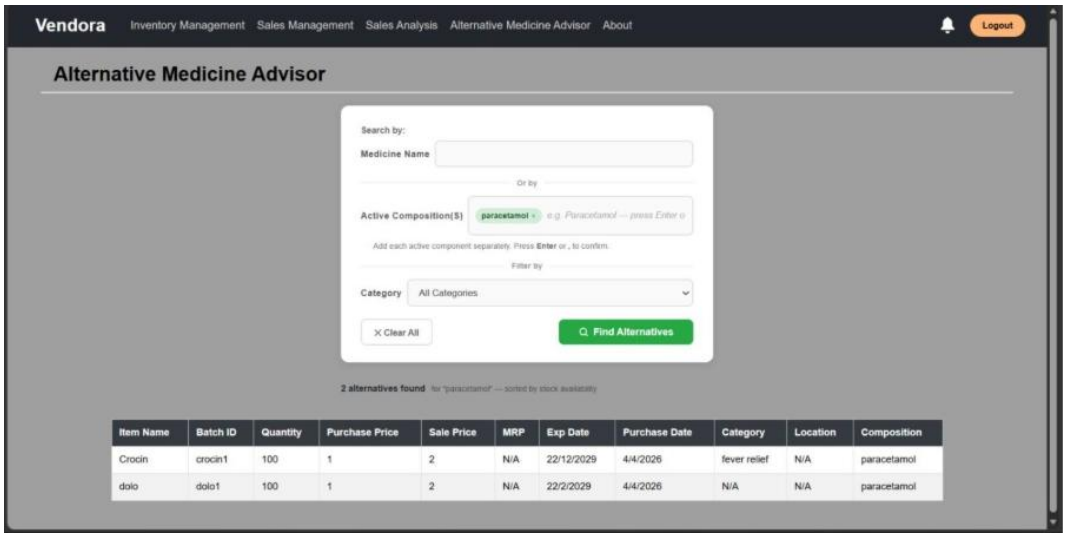


Fig – Alternative Medicine Advisor page

Purpose: To help Pharmacists find in-stock alternative medicines from their own inventory that share the same active composition(s) as a requested medicine.

Access: Restricted to Pharmacist accounts only. Non-pharmacist users and logged-out users are automatically redirected away from this page.

Input Fields

- **Medicine Name** — Text input with live autocomplete. Type at least 2 characters to see suggestions from your inventory.
- **Active Composition(s)** — Tag-style input with live autocomplete. Enter each active ingredient separately as a tag.
- **Category** — Dropdown filter. Populated automatically from the categories present in your inventory. Defaults to "All Categories".

Instructions:

1. **To Search by Medicine Name:** Click on the Medicine Name field. Select from autocomplete suggestions and enter to confirm.
2. **To Search by Active Composition(s):**
   i. Click on the **Active Composition(s)** field. Type an active component. Select from autocomplete suggestions. Press **Enter** or **,** (comma) to confirm each tag, or click a autocomplete suggestion to add it directly as a tag.
   ii. Repeat for additional ingredients. Each ingredient is added as a separate green tag.
   iii. Press the '**x**' button on the right end of the tag to remove it.
3. **Filter by Category (optional):**
   i. Click the Category dropdown. Select a category. The dropdown is populated dynamically from the categories in your inventory.

ii. You may use Category filter alone, or combined with a Name or Composition search.

iii. If only category is chosen, all records of that category are displayed.

4. Click **Find Alternatives** to search.
5. Click ✕ **Clear All** to reset all fields.

Note:

- **Medicine Name and Composition cannot be used together.** If both fields have values and you click Find Alternatives, an error message will appear. Clear one field before searching.
- Searching by **Category alone** (with Medicine Name and Composition left empty) is allowed — results will show all in-stock items in that category.
- The autocomplete suggestions for **Medicine Name** are drawn from medicine names already in your inventory.
- The autocomplete suggestions for **Active Composition(s)** are drawn from the composition data recorded in your inventory.
- To **remove a composition tag**, click the × button on the tag, or press **Backspace** when the text input is empty.
- The status bar above the results table shows the number of alternatives found and the search query used.
- Only **in-stock** items (Quantity > 0) are returned as results.

9. Sales Analysis Page

➢ Feature: Sales Trends

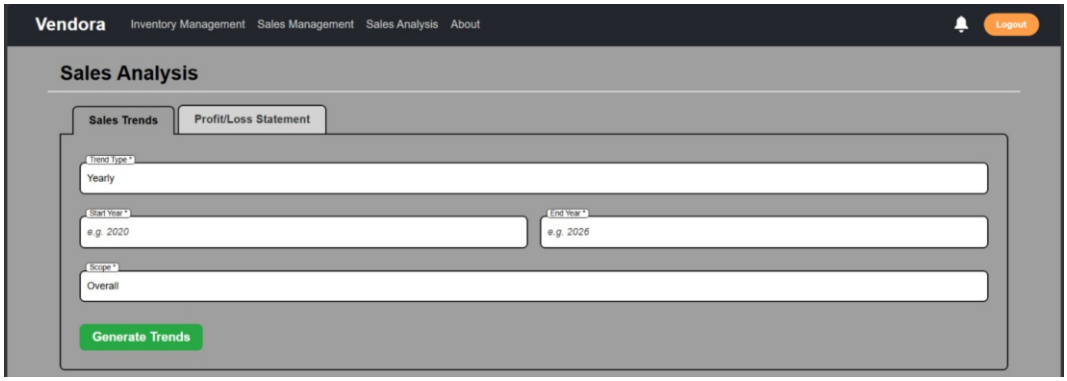


Fig – Sales Trend Tab

Purpose: The Sales Trends tab allows users to visualize profit trends over time based on selected filters such as duration, product scope, and time range.

Input Fields

- Trend Type: Select Yearly or Monthly analysis

- Date Range: Start and End duration based on trend type
- Scope: Overall or Specific Product
- Item Name: Required when Specific Product is selected

Instructions:

1. Select Trend Type
2. Enter Date Range
3. Choose Scope
4. Select Item (if required)
5. Click Generate Trends

Fig – Overall Sales Trend Output

Note:

- Line graph showing Total Net Profit over time
- Bar chart showing top-performing products
- Product-specific trend graph when filtered

➢ Profit/Loss Statement

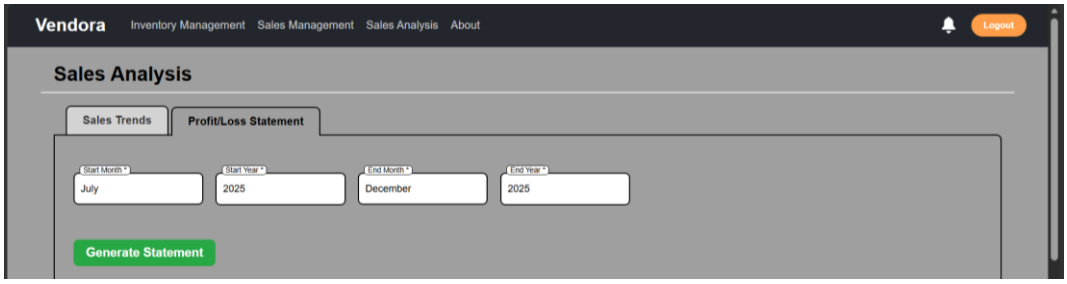


Fig – Profit/Loss Statement Tab

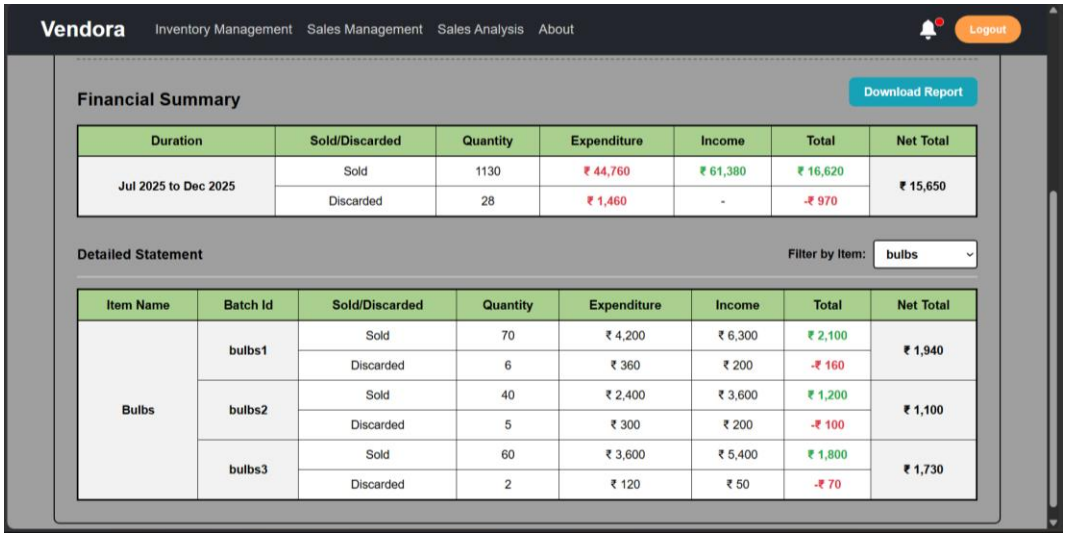
Purpose: Provides a financial summary of sold and discarded items within a selected period.

Input Fields:

- Start Month & Year
- End Month & Year

Instructions:

1. Select duration
2. Click Generate Statement
3. View summary and detailed product wise statements. The detailed statement can be filtered by item name.
4. Click on **'Download Report'** button to open print dialogue for the entire report.



Duration	Sold/Discarded	Quantity	Expenditure	Income	Total	Net Total
Jul 2025 to Dec 2025	Sold	1130	₹ 44,760	₹ 61,380	₹ 16,620	₹ 15,650
	Discarded	28	₹ 1,460	-	-₹ 970	

Item Name	Batch Id	Sold/Discarded	Quantity	Expenditure	Income	Total	Net Total
Bulbs	bulbs1	Sold	70	₹ 4,200	₹ 6,300	₹ 2,100	₹ 1,940
		Discarded	6	₹ 360	₹ 200	-₹ 160	
	bulbs2	Sold	40	₹ 2,400	₹ 3,600	₹ 1,200	₹ 1,100
		Discarded	5	₹ 300	₹ 200	-₹ 100	
	bulbs3	Sold	60	₹ 3,600	₹ 5,400	₹ 1,800	₹ 1,730
		Discarded	2	₹ 120	₹ 50	-₹ 70	

Fig – Proft/Loss Statement Output

Notes:

- Discarded items can also have refunds. The refund is stored as income.
- Net Total = Net sold profit/loss + Net discarded profit/loss
- Data is calculated from records stored in the database.
- For best print quality, in the print dialogue set orientation to landscape; page size to A4 and scale to 'Fit to printable area'(in windows these options are present in **More settings** dropdown).